

Date	16 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	2 Marks

Symbol	Name	Description
Oval / Rounded shape	External Entity	A person, organization, or system outside the project that sends or receives data (e.g., User, Google Gemini AI API).
Rectangle with numbered header	Process	An activity that transforms incoming data into outgoing data (e.g., 1.0 Authenticate User).
Rectangle (solid fill, no number)	Data Store	A place where data is held for later use (e.g., D1 Users DB, D2 Blogs DB, D3 Comments DB).
Labeled arrow	Data Flow	The movement of data between entities, processes, and data stores, labeled with what data is being passed.

```
graph TD
    User[External Entity: User Client / Postman] -- "↓ registration / login / blog / comment / AI requests | ↑ JWT, JSON responses" --> System
    subgraph System
        direction TB
        subgraph Row1
            direction LR
            B1["1.0 Authenticate & Authorize User  
validates credentials, issues JWT, enforces role Admin/Editor/Author/Reader via role middleware"]
            B2["2.0 Manage Blog Posts  
blogController.js - create/read/update/delete posts, drafts, publishing status, categories & tags"]
        end
        B1 <--> B2
        subgraph Row2
            direction LR
            D1["D1 — Users DB MongoDB"]
            D2["D2 — Blogs DB MongoDB"]
        end
        B1 <--> D1
        B2 <--> D2
        subgraph Row3
            direction LR
            B3["3.0 Manage Comments  
readers add comments; Editors/Admins moderate status: pending -> approved"]
            B4["4.0 AI Blog Generation & Summarization  
geminiService.js sends a topic generate or full content summarize to Gemini"]
        end
        B3 <--> B4
        subgraph Row4
            direction LR
            D3["D3 — Comments DB MongoDB"]
            API["External Entity: Google Gemini AI API"]
        end
        B3 <--> D3
        B4 <--> API
    end
    System -- "↓ structured JSON post / summary / comment returned to User" --> User
    User -- "External Entity: User Client receives final response" --> End[ ]
```

External Entity: **User (Client / Postman)**

↓ registration / login / blog / comment / AI requests | ↑ JWT, JSON responses

1.0 Authenticate & Authorize User
validates credentials, issues JWT, enforces role (Admin/Editor/Author/Reader) via role middleware

2.0 Manage Blog Posts
blogController.js - create/read/update/delete posts, drafts, publishing status, categories & tags

↓ ↑

D1 — Users DB (MongoDB)

D2 — Blogs DB (MongoDB)

3.0 Manage Comments
readers add comments; Editors/Admins moderate (status: pending → approved)

4.0 AI Blog Generation & Summarization
geminiService.js sends a topic (generate) or full content (summarize) to Gemini

↔ topic / content payload | generated post / summary text

D3 — Comments DB (MongoDB)

External Entity: **Google Gemini AI API**

↓ structured JSON (post / summary / comment) returned to User

External Entity: **User (Client)** receives final response

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